Document Control

Title			
Antibiotic Guidelines for Initial and Ongoing Management of Severe Sepsis and Septic Shock			
Author Dr Tom Lewis		**Author's job title** Consultant Microbiologist	
Directorate Diagnostics		**Department** Pathology	
Version	**Date Issued**	**Status**	**Comment / Changes / Approval**
0.1	Mar 11	Draft	First draft for consultation
0.2	Apr 11	Draft	Review of penicillin allergy recommendation in septic shock. Change to amoxicillin dose.
1.0	May 11	Final	Approved by Drug and Therapeutics Group on 12th May. Change to recommendation for management of septic shock and penicillin allergy.
1.1	May 11	Revision	Minor amendments by Corporate Affairs to update to latest template.
1.2	May 15	Revision	Email address and telephone numbers altered as per updated contact details, surviving sepsis guideline reference updated, clarification added: 2 sets of blood cultures required for pathology. Comment re: fulminant pneumonia altered to reflect current advice. Comment added re: checking previous microbiology samples. Guidance in case of MRSA clarified.
2.0	July 15	Final	Approved by Drug and Therapeutics Committee on 16th July and published on Bob.
2.1	Jun 18	Revision	Inclusion of initial management in ED guideline, update references, change of title to "initial and ongoing management of severe sepsis and septic shock". Approved at DTC 19th July.
2.2	Jul 21	Revision	Restructured guideline after discussion with working group. Extensive changes throughout clinical section. Addition of compliance table to compliance monitoring section with NG15 and NG51 criteria. Approved 3rd Aug 2021 IPDG.
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Lead Director Medical Director			
Superseded Documents			

Issue Date	Review Date	Review Cycle
August 2021	August 2024	Three years

Consulted with the following stakeholders:

- Antibiotic Working Group
- ED Governance
- Infection Prevention and Decontamination Group
- Consultant Physicians

Approval and Review Process

- Clinical Audit and Guidelines Group

Local Archive Reference

G:\ANTIBIOTIC STEWARDSHIP\Stewardship\Antibiotic policies\Archived policies

Local Path

Pharmacy\Antibiotic stewardship\Final policies\

Filename

Antibiotic guidelines for severe sepsis V2.2

Policy categories for Trust's internal website (Bob)	Tags for Trust's internal website (Bob)
Pharmacy	Infection, NEWS2, Microbiology

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1. Introduction

1.1. This document sets out Northern Devon Healthcare NHS Trust's best practice guidelines for initial management, appropriate microbiological investigation and antimicrobial prescribing in adult patients with suspected and confirmed sepsis of unknown origin.

1.2. This guideline applies to all adults except those who may have neutropenic sepsis, and maternity patients.

1.3. Non-compliance with this guideline may be for valid clinical reasons only. The reason(s) for non-compliance must be documented clearly in the patient's notes.

1.4. This guideline is primarily aimed at all prescribing teams but other staff (e.g. nursing staff, pharmacists) may need to familiarise themselves with some aspects of the guideline.

1.5. Implementation of this guideline will ensure that:

- Sepsis identified and managed in a timely and appropriate fashion
- There is a reduction in unnecessary antibiotic prescribing

2. Responsibilities

2.1. Responsibility for education and training lies with the Lead Consultant Microbiologist for Antibiotic Stewardship. It will be provided through formal study days and informal training on the ward.

2.2. The author will be responsible for ensuring the guidelines are reviewed and revisions approved by the Infection Prevention and Decontamination Group in accordance with the Document Control Report.

2.3. All versions of these guidelines will be archived in electronic format by the author within the Antibiotic Stewardship policy archive.

2.4. Any revisions to the final document will be recorded on the Document Control Report.

2.5. To obtain a copy of the archived guidelines, contact should be made with the author.

2.6. Monitoring of implementation, effectiveness and compliance with these guidelines will be the responsibility of the Lead Clinician for Antibiotic Stewardship. Where non-compliance is found, the reasons for this must have been documented in the patient's medical notes.

Role of Antibiotic Working Group (AWG)

2.7. The AWG is responsible for:

- Leading antibiotic guideline development and review within Northern Devon Healthcare Trust
- Involving all relevant stakeholders in guideline development and review

3. Contacts

3.1. Contact numbers:

- Microbiologist — Bleep 193. Via switchboard out of hours.
- Antibiotic Pharmacist — Bleep 029 (Mon-Fri only)

4. Management of Sepsis

4.1. See table of contents for appendices

4.2. In Feb 2016 an International Panel of experts recommended a new definition of Sepsis and published new guidelines for the management of this common condition. The new recommendations define sepsis as 'life-threatening organ dysfunction due to a dysregulated host response to infection'. Septic shock is defined as 'a subset of sepsis in which particularly profound circulatory, cellular, and metabolic abnormalities substantially increase mortality'.

4.3. This new definition is important as the term sepsis now implies an immediately life threatening condition. Previously the term sepsis has frequently been used interchangeably with the term infection, for example, when the patient has a mild fever or blood test abnormalities.

4.4. Clarity of diagnosis in these patients is important to allow clear handover between teams and to allow accurate recording of hospital admission data. A link to the current guidelines can be found at http://jamanetwork.com/journals/jama/fullarticle/2492881

4.5. NICE guidance for management of sepsis recommends that a structured set of observations should be used as the basis of early detection. A variety of scoring systems have been proposed, but there is ongoing debate about how these perform in a real-life setting. There is general agreement that SIRS criteria are insufficiently specific, and local data suggests that a promotion of SIRS based sepsis criteria led to a 50% increase in antibiotic use in the emergency department. NICE guidance recommends consideration of an early warning score in an acute hospital setting. The Trust currently uses the National Early Warning Score (NEWS2) for identification of patients at risk of deterioration, and this is a well validated way of identifying patients who are unwell. Doctors are used to working with it, and recognise its limitations. At sepsis steering board meetings, it was felt that NEWS2 should identify most patients with sepsis, and that there would be an opportunity cost to introducing a new scoring system that was specific for sepsis. This would need significant resources for ongoing training, and has the potential to detract from assessment of other conditions. It was noted that a patient who triggered on the qSOFA score would have a NEWS2 of at least 5. The Trust NEWS2 cut-off score for patients at serious risk of deterioration is 7, and it was therefore decided that red-flag sepsis would be defined in this Trust as suspected infection with NEWS2 of 7 or more. Red-flag sepsis requires immediate action. NICE guidance recommends patients in a moderate risk group should be reviewed by a senior clinician within 3 hours of presentation. A category of amber flag suspected sepsis was created to include patients with an EWS between 4 and 6. This is in line with the Trust's deteriorating patient moderate risk category. Patients in this group need close monitoring and use of diagnostics to ascertain the likelihood and severity of possible infection.

4.6. It was also noted that patients with red flag sepsis should have antibiotics within an hour. It is hard to do this, and the main reasons that were identified for this were 1) difficulty in making a diagnosis with an hour and 2) complex antibiotic protocols that require an accurate diagnosis. It was also felt that SIRS based definitions of sepsis had pushed people into treating infections rapidly when this did not marry with the clinical judgement of severity. This may have led to a downplaying of the potential severity of sepsis in some cases. Conversely, there was probably a knee-jerk prescribing of antibiotics in some situations when SIRS criteria had triggered, but evidence of infection may have been lacking. In other words, there is a behavioural cost of scoring systems that lack specificity.

4.7. It was also noted that presentation of sepsis to the emergency department is often non-specific. This results in the final diagnosis differing from the working diagnosis made on presentation in up to 50% of cases. It is unlikely that this could be improved, as there is not sufficient time to obtain the relevant information in which to make a fully informed decision. The result of this can be patients treated on inappropriate antibiotic protocols, but also inappropriate diagnoses that can 'stick' to the patient throughout their subsequent admission.

4.8. A further finding in case analysis was that the definitions of sepsis rely on infection being recognised in the differential diagnosis. It was found that people have different understandings of how to make this assessment. A common misconception was that patients needed to be febrile to be septic. Instead, case study suggested there should be an emphasis on fever history (particularly rigors), changes in mental state, and any other concern about acute deterioration raised by carers. These findings are consistent with recommendation in NICE guidance.

4.9. It was concluded that an ideal sepsis pathway in ED would comprise:

a) Clear pointers on when to consider infection

b) A triage scoring system that was easy to use

c) Clear guidance on how to manage patients in different triage risk groups

d) In patients with red flag sepsis, antibiotic protocols must cover all common organisms (Group A streptococcus, S. aureus, Pneumococcus, E. coli, meningococcus) and be easy to administer, with few contraindications or caveats. Ideally antibiotic choices should be agnostic to actual diagnosis, should cover all common pathogens, and should not be in other antibiotic protocols (as this forces a subsequent review of management plan and diagnosis). Three possibilities were felt to fit this criteria : ertapenem; fosfomycin and chloramphenicol.

These conclusions were supported with some audit evidence.

1. 100 ED notes of patients were reviewed. Of these patients, only five had an EWS of 6 or more. Of these five patients, three has presumed infection, two with an obvious chest focus. All received antibiotics. There were 9 patients with an EWS (audit done prior to launch of NEWS2 score) of 3, 4 or 5, and infection was considered as a cause for their presentation in four cases. ED consultant review suggested that these patients were not overtly 'septic' and could be monitored pending definitive management.
2. 15 cases of bacteraemia detected in ED taken blood cultures were reviewed. 5 cases would have classified as 'red flag'; 8 cases as 'amber flag' and 2 cases 'green flag'. These classifications were felt to reflect accurately the likely severity and subsequent evolution of the case. In only 3 cases of urosepsis was the diagnosis felt to be clear on initial presentation. Patients generally presented with non-specific syndromes of 'being unwell', often with a history of sweats or chills (10 patients). In the other patients, the next most common presenting symptom was change in mental state (3 patients). Two patients had very non-specific presentations (vomiting, general malaise).
3. A random sample of 12 patients (out of a total of 154) who had a blood cultures taken in October 2016 was reviewed. 5 patients had an EWS of 6 or more. 4 patients had an EWS of

3-5. Extrapolating, we would expect about 10 cases of red flag sepsis per week, and 6 cases of amber flag sepsis. No patients had an antibiotic within an hour of presentations. In patients with an EWS of 6 or more, average time to antibiotics was nearly 2 hours. Reviewing subsequent antibiotics, there was a total of 66 days of antibiotic exposure; including 2 days of carbapenem exposure, 5 days of tazocin exposure and 13 days of co-amoxiclav exposure. 2 cases were subsequently diagnosed with other diagnosis (viral LRTI; headache).

4. In 40 patients with urosepsis due to E coli with bacteraemia detected in the ED, only 7 had 2 sets of blood cultures taken. In patients with 2 sets taken, the episode positivity rate was 19%. In those with only 1 set taken the episode positivity rate was 8%. 7 patients had no urine sent. 14 patients had a urine sent after discharge from ED. Only 5/14 grew E. coli, compared with 16/19 if the urine was sent by the GP prior to admission. In about 50% of the cases were clinical symptoms consistent with UTI recorded on the request form. In the other cases, either non-specific, or other diagnoses were recorded (eg. LRTI).
5. In a review of the utility of 112 positive blood cultures, the culture led to identification of an unknown focus in 21 cases and led to a significant change of diagnosis in 14 cases. Antibiotics were changed as a consequence of results in 45 cases (gram stain in 9 cases, organism identification in 17 cases, sensitivity results in 19 cases)
6. Isolates of *E coli*, Group A *Strep* and *S. aureus* were tested against chloramphenicol, fosfomycin and ertapenem. Sensitivity was 100% for all antibiotic/organism combinations.

5. Monitoring Compliance with and the Effectiveness of the Guideline

Suggested audit criteria

- EWS in patients prescribed chloramphenicol
- Final diagnosis in patients given chloramphenicol
- Follow on antibiotic choice

Process for Implementation and Monitoring Compliance and Effectiveness

5.1. Incidents involving sepsis should be reported according to the Trust's Incident Reporting Policy. Critical incident reports relating to sepsis will be collated by the Antibiotic Pharmacist. Results will be reported on an annual basis to the Drug and Therapeutics Group.

5.2. Statement of compliance with NICE NG51

Relevant section	NICE statement	Guidance
1.1.2.	Take into account that people with sepsis may have non-specific, non-localised presentations, for example feeling very unwell, and may not have a high temperature.	Take into account that people with sepsis may have non-specific, non-localised presentations, for example feeling very unwell, and may not have a high temperature.
1.1.8	Consider using an early warning score (NEWS2 has been endorsed by NHS England) to assess people with suspected sepsis in acute hospital settings	Based on hospital deteriorating patient guideline (NEWS2)

1.6.1	carry out a venous blood test for the following: — blood gas including glucose and lactate measurement — blood culture — full blood count — C-reactive protein — urea and electrolytes — creatinine — a clotting screen	BEFORE ANTIBIOTICS, carry out a venous blood test for the following: • blood culture (10ml of blood into each of an aerobic and anaerobic bottle) • blood gas including glucose and lactate measurement • full blood count; CRP; Renal profile; Liver profile; INR Ideally, a second set of blood cultures should be taken before antibiotics
1.7.13	Follow the recommendations in NICE's guideline on antimicrobial stewardship: systems and processes for effective antimicrobial medicine when prescribing and using antibiotics to treat people with suspected or confirmed sepsis.	See below (NG15)

5.3. Statement of compliance with NICE NG15:

Relevant section	NICE statement	Guidance
1.1.29	For patients who have non-severe infections, consider taking microbiological samples before making a decision about prescribing an antimicrobial, providing it is safe to withhold treatment until the results are available.	Differentiation between Amber and Red flag sepsis
1.2.7	When evaluating a new antimicrobial for local use and for inclusion in the local formulary, take into account: • the need for the new antimicrobial • its clinical effectiveness • the population in which it will be used • the specific organisms or conditions for which it will be used • dose, dose frequency, formulation and route of administration • likely tolerability and adherence • any drug interactions, contraindications or cautions • local rates and trends of resistance • whether use should be restricted and, if so, how use will be monitored • any additional monitoring needed • any urgent clinical need for the new antimicrobial • any plans for introducing the new antimicrobial.	See principles for choosing ED sepsis antibiotic; and epidemiological data above.

6. Equality Impact Assessment

6.1. The author must include the Equality Impact Assessment Table and identify whether the policy has a positive or negative impact on any of the groups listed. The Author must make comment on how the policy makes this impact.

Table 1: Equality impact Assessment

Group	Positive Impact	Negative Impact	No Impact	**Comment**
Age	X			Separate guidance for paediatrics
Disability			X	
Gender			X	
Gender Reassignment			X	
Human Rights (rights to privacy, dignity, liberty and non-degrading treatment)			X	
Marriage and civil partnership			X	
Pregnancy		X		Some treatment advice may harm the unborn foetus, discuss on a case-by-case basis with Obstetricians and Pharmacy for advice.
Maternity and Breastfeeding		X		Some treatments may be excreted in breast milk. Discuss on a case-by-case basis with Paediatricians and Pharmacy for advice.
Race (ethnic origin)			X	
Religion (or belief)			X	

7. References

7.1. NICE guideline 51. Sepsis: Recognition, Diagnosis and Early Management. July 2016, updated 2017, 2019.

7.2. Surviving Sepsis Campaign. 2016. International Guidelines for Management of Sepsis and Septic Shock 4th edition.

7.3. Patients at risk of deterioration policy v4.0. NDHT 2019.

7.4. Supporting Evidence Commentary:

- Relevant NICE guidance consulted :
  - NG51. 2016. https://www.nice.org.uk/guidance/ng51/resources/sepsis-recognition-diagnosis-and-early-management-pdf-1837508256709 (accessed 16th March 2021)
  - Antimicrobial stewardship: systems and processes for effective antimicrobial medicine use. NICE guideline [NG15]Published date: 18 August 2015 https://www.nice.org.uk/guidance/ng15 (accessed 16th March 2021).
- Other relevant national guidance
  - Review of NEWS2 guidance https://www.rcplondon.ac.uk/projects/outputs/national-early-warning-score-news-2 (accessed 16th March 2021).. In a patient with a confirmed or suspected infection, or at risk of infection, a NEW score of 5 or more should raise the suspicion of sepsis and prompt an urgent clinical response by a clinical team competent in the management of sepsis and higher-dependency care.
- Local guidance
  - Align Sepsis protocols with deteriorating patient protocol https://www.northdevonhealth.nhs.uk/wp-content/uploads/2019/05/Patients-at-Risk-of-Deterioration-Policy-V4.0.pdf (accessed 16th March 2021). Note change from previous guidance. Red flag sepsis was EWS 6, now changed to 7. Include single score of 3; or EWS 5-6 as amber.
  - Note deteriorating patient protocol does not include room for clinical judgement when elevated baseline EWS. The need for clinical judgement has been included.
- Other antibiotic guidelines
  - Amalgamated sepsis protocols from ED and inpatient.
  - Use of sepsis antibiotic continued in ED due to positive feedback about ease of use.
  - Outside the ED, choice of antibiotic to be determined by clinical assessment. The reason for this is that there is more history, examination and investigation data on which to base this assessment
  - Separated infection of unknown origin protocol
  - It has been confusing including this in the sepsis protocols and have not been used widely / consistently.
- Antibiotic choice. Principles of ED sepsis antibiotic remain :
  - Easy to administer
  - No common contraindications
  - Likely activity in common infections causing sepsis:

- Pneumonia
- UTI
- Cholangitis
- Cellulitis
- Meningitis

- Not in other hospital antibiotic guidelines
- Low rates of resistance in common pathogens
- Low risk for side effects, including *C difficile*

8. Associated Documentation

- Incident reporting policy
- Antibiotic guidelines for management of severe sepsis and septic shock
- Antibiotic prescribing policy
- Penicillin allergy policy

9. Appendix 1: Sepsis: Initial management

9.1. Name of guideline on app

Sepsis – initial management

9.2. Location on app

Infection
Secondary Care
Adult treatment
Severe Sepsis and Septic Shock
Sepsis – Initial Management

9.3. Suspected Red Flag Sepsis (NEWS2 7+)

Red flag sepsis is defined by a NEWS2 of 7 or more in the presence of suspected infection

Suspect infection if :

- History of rigors or fevers
- No other explanation for clinical presentation

Take into account that people with sepsis may have non-specific, non-localised presentations, for example feeling very unwell, and may not have a high temperature.

See link to NICE sepsis recognition guidance (requires internet connection):
https://www.nice.org.uk/guidance/ng51/resources/table-1-risk-stratification-tool-for-adults-children-and-young-people-aged-12-years-and-over-with-suspected-sepsis-2551487005

- **Investigations**

BEFORE ANTIBIOTICS, carry out a venous blood test for the following:

- blood culture (10ml of blood into each of an aerobic and anaerobic bottle)
- blood gas including glucose and lactate measurement
- full blood count; CRP; Renal profile; Liver profile; INR

Ideally, a second set of blood cultures should be taken before antibiotics

- **Initial Treatment in the Emergency Department (NOT FOR INPATIENT WARD USE)**

Chloramphenicol 1g IV (bolus)

Do not treat according to possible diagnosis. This is often inaccurate on initial presentation.

Do not give another dose of chloramphenicol under any circumstances

- **Initial treatment: Outside the Emergency Department (incl. inpatient wards)**

Treat immediately according to specific infection protocol

- **Follow-up**

Patients must be reviewed within 3 hours.

At the time of review a decision should be made to either

- Not prescribe further antibiotics if sepsis no longer felt to be a likely diagnosis
- Prescribe antibiotics according to the likely focus of infection
- Prescribe antibiotics according to the "infection of unknown origin guideline"

- **Other Relevant Guidelines**

Severe Sepsis and Septic Shock Guidelines, and Septic Shock Algorithm

- **Organisms and Sensitivities**

Group A Streptococcus

S. aureus

Pneumococcus

E coli

Meningococcus

- **Version Control**

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9.4. Suspected Amber Flag sepsis (NEWS2 5-6)

Amber flag sepsis is defined by a NEWS2 of 5 or 6, OR

A single parameter in the NEWS2 with a score of 3;

AND in the presence of suspected infection

Suspect infection if :

History of rigors or fevers

No other explanation for clinical presentation

Take into account that people with sepsis may have non-specific, non-localised presentations, for example feeling very unwell, and may not have a high temperature.

Blood tests are not part of the initial assessment of sepsis.

See NICE sepsis risk stratification tool for help with deciding between red, amber and green risk criteria: https://www.nice.org.uk/guidance/ng51/resources/table-1-risk-stratification-tool-for-adults-children-and-young-people-aged-12-years-and-over-with-suspected-sepsis-2551487005

- **Investigations**

BEFORE ANTIBIOTICS, carry out a venous blood test for the following:

- blood culture (10ml of blood into each of an aerobic and anaerobic bottle)
- blood gas including glucose and lactate measurement
- full blood count; CRP; Renal profile; Liver profile; INR

Ideally, a second set of blood cultures should be taken before antibiotics

- **Initial treatment in all locations**

Do not give antibiotics straightaway

Review within 3 hours with results of initial investigations

Treat as appropriate, using specific infection protocol

9.5. Suspected Infection with NEWS2 of 4 or less

In patients with suspected infection with a NEWS2 of 4 or less, urgent prescribing of an antimicrobial is not necessary.

- **Investigations**

  - Carry out investigations as directed by history and examination
  - Blood cultures are rarely useful in this context unless:

- there is a history of rigors
- endocarditis, or other endovascular infection, is possible

- **Initial treatment in all locations**
  - Do not give antibiotics until a definitive diagnosis has been made
  - Treat as appropriate, using specific infection protocol

10. Appendix 2: Septic Shock Refractory to Initial management

10.1. Name of guideline on app

Septic Shock Refractory to Initial management

10.2. Location on app

Infection
Secondary Care
Adult treatment
Severe Sepsis and Septic Shock
Septic shock refractory to initial management

10.3. This guideline refers to patients who:

- Have a persistently high NEWS2 score (7 or more) with suspected infection
- No evidence of improvement in the first few hours after initial management using the sepsis protocol
- Lack alternative explanation for their ongoing physiological state

10.4. These patients should be reviewed by the ICU team.

- In the absence of a clear focus of infection then ensure blood cultures and urine have been sent.

10.5. Start antibiotic treatment:

- **Meropenem 500mg QDS IV**

11. Appendix 3: Infection of Unknown Origin NEWS2 5+

11.1. Name of guideline on app

Infection of Unknown Origin NEWS2 5+

11.2. Location on app

Infection
Secondary Care
Adult treatment
Severe Sepsis and Septic Shock
Infection of Unknown Origin NEWS2 5+

11.3. This guideline refers to patients who have:

Suspected infection with no clear focus AND
NEWS2 of 5 or more

11.4. Investigations

- Ensure at least 2 sets of blood cultures have been taken, preferably whilst OFF antibiotics - for at least a few days but ideally 2 weeks:
  - Fill each bottle to 10mL
  - Take different sets from different sites if possible and write on form where each set was taken from
- Ensure a urine sample has been sent, preferably whilst OFF antibiotics – for at least a few days:
  - Preferably MSU
  - If sending catheter sample - from freshly inserted catheter if possible

11.5. Initial Treatment for first 24 hours

- **Gentamicin IV (as per protocol)**
  - Single dose

Plus

- **Amoxicillin 1g QDS IV**

Or, if penicillin allergy, then consider instead adding:

- **Vancomycin IV (as per protocol)**
  - Note that gentamicin covers most common pathogens apart from *Streptococci* and *Meningococci*.
  - It may be appropriate to withhold vancomycin and see how the patient responds to gentamicin alone.

11.6. Ongoing Treatment after first 24 hours

- These patients MUST be reviewed the following day.
- Generally, the patient should have improved with a NEWS2 less than 5.
- If the NEWS2 remains 5 or more, without a clear explanation, then discuss with the on call microbiologist.
- With the benefit of time, further review, and results of investigations, it should be possible to either
  - Stop antibiotics
  - Switch to the antibiotic guideline appropriate to the refined diagnosis
  - Switch to the suspected infection with no clear focus and NEWS2 4 or less protocol

12. Appendix 3: Suspected Infection No Clear Focus NEWS2 4 or less

12.1. Name of guideline on app

Suspected Infection No Clear Focus NEWS2 4 or less

12.2. Location on app

Infection
Secondary Care
Adult treatment
Severe Sepsis and Septic Shock
Suspected Infection No Clear Focus NEWS2 4 or less

12.3. Diagnosis Help Notes

The cause of a pyrexia of unknown origin can generally be split into:

- infective (25%)
- malignant (25%)
- rheumatological (25%)
- unknown (25%)

Similiar proportions of aetiologies probably account for raised CRP and raised WBC without a clear focus.

Common reasons for an INFECTION of unknown origin

- Urinary catheters
- Infected haematoma (particularly after surgery)
- Clostridium difficile
- Endocarditis (especially if any history of valve issues)
- Discitis
- Septic arthritis
- Sub-phrenic collections
- Viral illnesses

12.4. Initial Management Approach

It is paramount that these patients are not given antibiotics

Giving antibiotics at this stage can severely compromise onward management

A simple initial approach does not usually entail much more than:

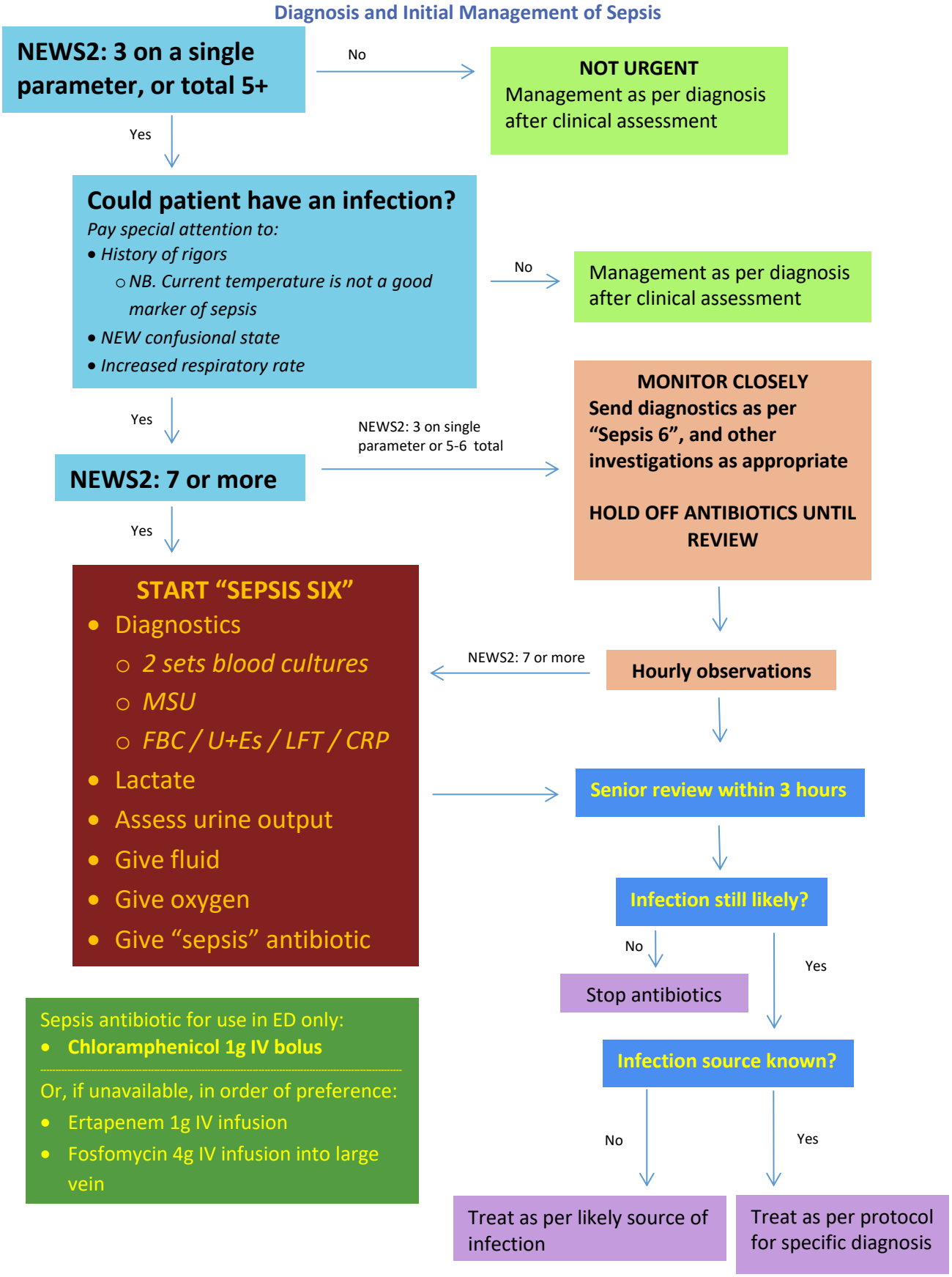
- Taking a good fever history, particularly:
  - When last feeling well?
  - Any rigors?
  - Any night sweats?
  - Rash history
  - Antibiotic history
  - Any possible immunocompromise (including >6/52 steroids)?
  - Any prosthetic material? (implants, joints, teeth, mesh, wires, plates)?
  - Other close contacts well?
  - New sexual partners? Or suspected STIs in regular sexual partners?

    - Travel / hobbies / pets / occupations
- Well taken blood cultures
    - At least 10ml of blood in each of 2 aerobic and 2 anaerobic bottles
    - Off antibiotics for at least 2 weeks
        - Call to discuss with microbiologist if recent antibiotic courses finishing within past 2 weeks, for best approach to sampling
- Basic imaging
    - Contrasted CT abdomen may be necessary to identify collections and lymphadenopathy, if no cause identified and there is ongoing suspicion of infection

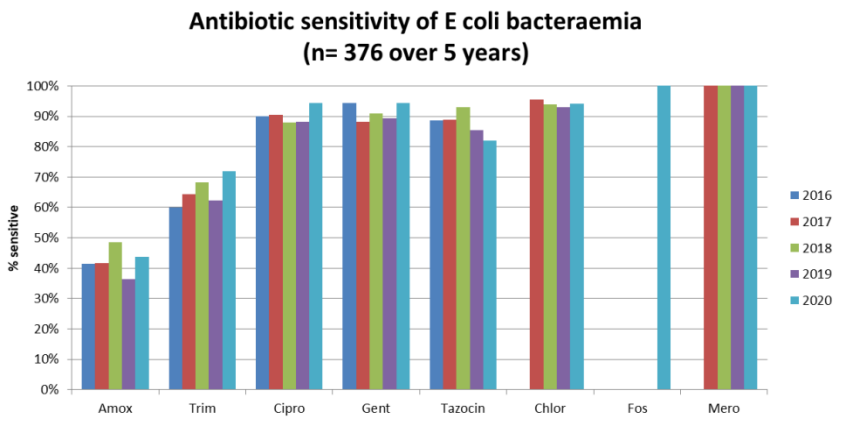
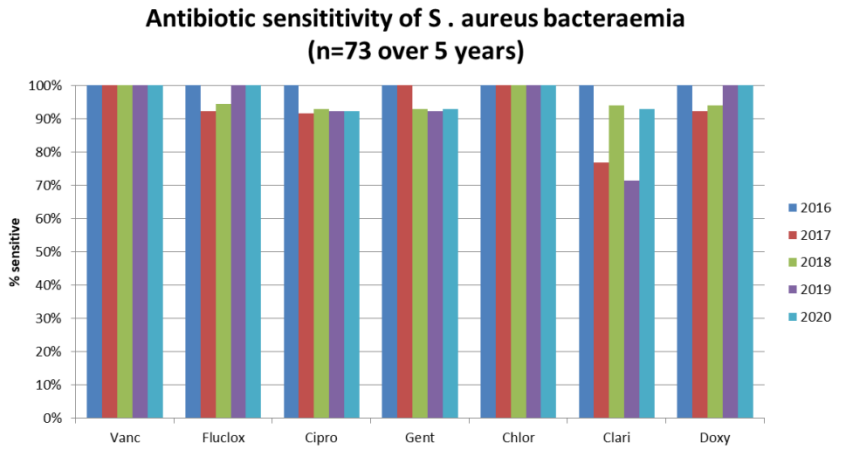
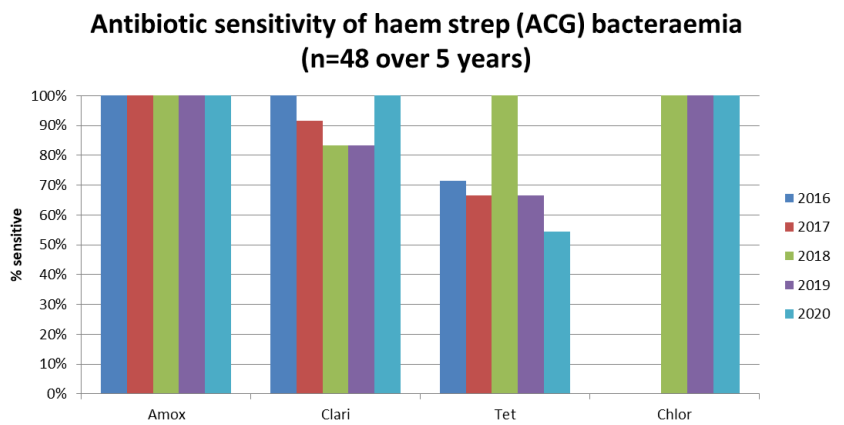
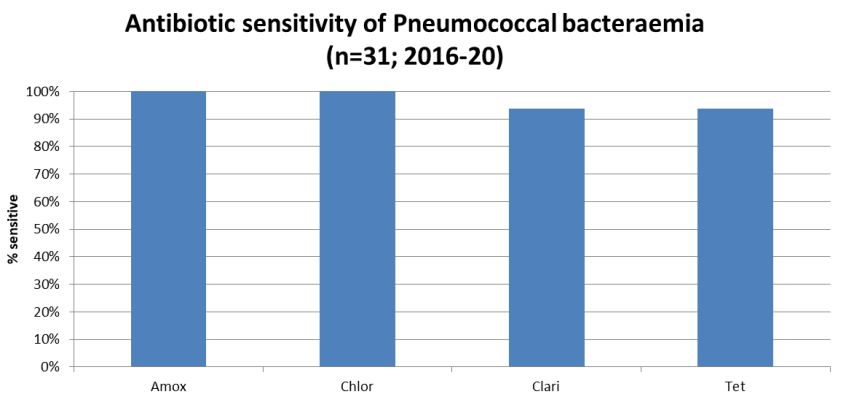
12.5. Ongoing Suspicion of Infection with Negative Initial Investigations

- Consider an echo
- Consider serology for viral and other infection, as directed by risk factors, history, examinations and investigations. Serology that is nearly always indicated in this situation:
    - EBV
    - CMV
    - HIV
    - Syphilis
- The likelihood of an infectious aetiology decreases as symptoms persist without a diagnosis
-

13. Appendix 4 - Algorithm for management of sepsis in the ED



14. Appendix 5: Supporting Information

E coli	
S. aureus	
Gp ACG strep	

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